13.6.2 Emerson's Enhancement Effect

Emerson modified his first experiment by supplying shorter wavelength of light (red light) along with longer wavelength of light (far red light). He found that the monochromatic light of longer wavelength (far red light) when supplemented with shorter wavelength of light (red light) enhanced photosynthetic yield and recovered red drop. This enhancement of photosynthetic yield is referred to as Emerson's Enhancement Effect (Figure 13.8).

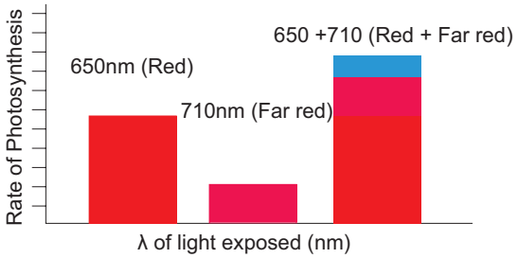


Figure 13.8: Emerson's Enhancement Effect

Photosynthetic rate at far red light (710 nm) = 10

Photosynthetic rate at red light (650 nm) = 43.5

Photosynthetic rate at red + far red (650 + 710 nm) = 72.5 (Enhancement effect).

13.6.3 Hill's Reaction

R. Hill (1937) isolated chloroplasts and when they were illuminated in the presence of suitable electron acceptors such as ferricyanide, they were reduced to ferrocyanide and oxygen is evolved. Hill's Reaction is now considered to be equivalent to Light Reaction.

Conclusions of Hill's Reaction:

1. During photosynthesis oxygen is evolved from water.
2. Electrons for the reduction of CO_2 are obtained from water.
3. Reduced substance produced, later helps to reduce CO_2

$$2H_2O + 2A \longrightarrow 2AH_2 + O_2$$

A is the Hydrogen acceptor, the common *in vitro* hydrogen acceptors are ferricyanide, benzoquinone and Di Chloro Phenol Indole Phenol (DCPIP).

13.7 Modern Concept of Photosynthesis

Photosynthesis is an Oxidation and Reduction process. Water is oxidised to release O_2 and CO_2 is reduced to form sugars. The first phase requires light and is called **light reaction or Hill's reaction**.

1. Light reaction: It is a photochemical reaction whereas dark reaction is a thermochemical reaction.

Solar energy is trapped by chlorophyll and stored in the form of chemical energy (assimilatory power)as ATP and reducing power NADPH + H^+. NADPH + H^+ alone are known as **reducing powers**. This reaction takes place in thylakoid membrane of the chloroplast. Oxygen is evolved as a result of splitting of water molecules by light.

Light reaction is discussed in two phases:

i. Photo-oxidation Phase:

- Absorption of light energy.
- Transfer of energy from accessory pigments to reaction centre.
- Activation of Chlorophyll 'a' molecule.

ii. Photo Chemical Phase:

- Photolysis of water and oxygen evolution
- Electron transport and synthesis of assimilatory power.

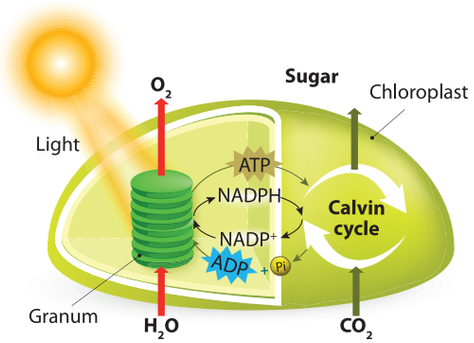


Figure 13.9: Light and Dark Reaction

2. Dark reaction (Biosynthetic phase): Fixation and reduction of CO_2 into carbohydrates with the help of assimilatory power produced during light reaction. This reaction does not require light and is not directly light driven.

XIth_BIO-BOTANY_Ch_13-EM.indd 247 5/17/2020 11:28:53 PM